

Missing Temperature in the Salish Sea

An analysis of basic regression methods, and their use in prediction temperatures within kelp forests.

Ahrial Young & Carter Webb

Introduction

Kelp in the Salish Sea is of utmost cultural, recreational, and ecological importance. The canopy of these forests are natively made up of Bull Kelp, scientifically named *Nereocystis luetkeana*. These forests have been present in the region for time immemorial, however, recent declines in the region, and around the world, have experts worried about the future of these ecosystems. It is well known that effects of temperature can severely affect the health of these forests, one of Washington State's most important species. Temperatures greater than 16°C can lead to deterioration of tissues and reduced fitness [3][4][5]. As research matures, it has become increasingly evident that environmental parameters, like temperature, can be highly variable across localized spatial dimensions, especially in marine estuaries, such as the Salish Sea. This can make it hard to predict just how severely a given forest will be affected by temperature fluctuations across the Salish Sea.

Recently, monitoring efforts to track the health of kelp forests across the Salish Sea have become more rigorous. In an effort to better understand trends here in the Salish Sea, large numbers of organizations have begun monitoring the state's kelp in a variety of ways. While these efforts are impressive in nature, field work can often leave something to be desired when it comes to consistency. Whether it be equipment failure, uncooperative weather, or lack of resources, temperature can oftentimes go unmonitored or be unreliable. For this reason, there is often the need to use alternative data sets for quality control or for direct imputation. With these methods a reliable and continuous data set can be created allowing for more confident analysis of kelp area/health and temperature relationships. With this in mind, this work uses k-nearest neighbors (kNN) regression to evaluate the effectiveness of imputing missing temperature values within local monitoring datasets. The nature of our data exhibits strong spatial and temporal variation across the Salish Sea, making them well suited for an introductory analysis using kNN by leveraging similarities between nearby observations within the sample space. The data set we are bringing in to assist comes from the Washington State Department of Ecology (WDOE) monitoring stations and provides a network of temperature observations and are distributed throughout the Northwest Straits Commission (NWSC) kelp monitoring sites. The spatial relationship between these two datasets, shown in Figure 1, allows us to make meaningful temperature estimates from nearby WDOE stations at NWSC kelp monitoring locations.

Data and Data Management:

Our data processing steps followed a simple, yet powerful outline. Relying on logical and scientifically sound decisions, we were able to distill a disordered and very large data set, down to one that could be efficiently used for our needs. We began with two datasets made of 3 tables, or relations. The first was that of a Washington Department of Ecology (WDOE) dataset, corresponding to Water Parameter and Sediment Monitoring efforts sourced from the Environmental Information Management (EIM) system within the WDOE website. This data spans 1999 - 2024 and represents a network of survey sites across Puget Sound where data on environmental parameters like temperature, dissolved oxygen and light attenuation are taken throughout the water column. A subset of WDOE - EIM sites were chosen due to their proximity to sites we wished to conduct imputation on. Those sites bring us to the second data set which consisted of two Northwest Straits Commission (NWSC) - Marine Resource Committees (MRCs) data tables, one which contained time series records of surveys conducted via kayak and another which had the housed the Latitude and Longitude coordinates of the sites which are monitored regularly.

To start, we identified the shared attributes that would be needed to run a regression model. Then, realising both datasets were made of time series data, it was determined that a singular fact sheet would best house our cleaned data. From there the following relational model was developed,

fact_table_clean(lid, date_time, name, parameter, result, result_units, depth, depth_units, lat, lon, NWSC_Max)

This task would entail removing some 20+ attributes from the WDOE dataset being and a further 7 from the NWSC survey datatable. From there, we were ready to begin doing more than just simplification.

During the 10 years of NWSC - MRC surveys, temperature monitoring methodology has changed. More specifically, in the initial years, only one Sea Surface Temperature (SST) was taken during a survey, subsequently, two SSTs were taken, one on the shoreside of the kelp forest, and one on the deep side; more recently, 4 temperatures have been taken, 2 on the shore side, 2 on the deep side. We began by seeking a single value representation of the temperature for any given survey at any given site. We accomplished this task by averaging temperature across their respective sites, then across the whole bed.

Finally, the last step in processing for the NWSC dataset was to combine the latitude and longitude coordinates with the surveys on the site id, then renaming it as lid on the fact_sheet. On the other hand, we took to preparing our WDOE data set by reducing it in scope, both in time, the dates we cared most about, 2014 - 2024, in depth, bull kelp grows 0.5 - 10 meters, and lastly, in parameter space, we aim to focus on

temperature in this project. From there we could do some simple renaming to align the attributes of the two cleaned datasets and prepare them for concatenation.

Exploratory Data Analysis (EDA):

Once concatenation was complete, we took to exploratory data analysis. To start, we wanted to ensure we had selected the scope properly. By calling the ranges of our 3 of variable attributes, date, depth, and temp, we got a better sense of the data. As we would expect, our depth ranges were SST, 0m, for NWSC and the range of Bull Kelp Growth, 0.5 - 10m, for WDOE. Similarly we had set our date ranges to match, with a bit of a buffer on the WDOE data to allow any regression analysis to search 2014 for data if necessary. The real illuminating feature, seen below in the table, is the difference in temperature ranges. This being the variable we wish to predict, it is important to notice that the temp range for WDOE, our training data set, is about 1.5 times as big as our target data set, NWSC. In practice, this means for any analysis we are doing, our training data set will likely be biased towards the lower end of the temperature spectrum. As a byproduct of having a depth range deeper in the water column, this isn't all that surprising to see this, however, though it should be kept in mind going forward.

Dataset	Date Range (Year)	Depth Range (m)	Temp Range (°C)
WDOE - EIM	2014 - 2024	0.5 - 10	5.4 - 20.9
NWSC - MRC	2015 - 2024	0.0 - 0.0	9.7 - 20.0

Next we needed to review the continuity of our data and give us a better visual understanding of the sites we were imputing at. This was best accomplished with a series of stacked line graph plots showing the temperature at each depth for every location id (lid) we had remaining in our data. A snapshot of the larger line graph can be seen in figure 2.

With this information we were able to move forward with quantifying the nature of our data set by observing the total observations, missing data we wished to impute (applicable only to NWSC), as well as mean and standard deviation. All of which can be seen in the table below. Knowing more about our data set we are able to see that the WDOE - EIM data set, not only has magnitudes more information, it generally is much colder and more variable. This further supports our suspicion that we may see inherent bias in our projections when using kNN alone.

Dataset	Total Observations	Missing Data	Mean Temp (°C)	STD
WDOE - EIM	16,451	NA	10.6	2.17
NWSC - MRC	119	18	13.48	1.99

Modeling Results:

kNN is used to address the missing temperatures in our kelp dataset by predicting those missing values using the average of the k number of nearest data points in the sample space. With this in mind, we must recall that this data does not have an inherently linear relationship and is highly collinear, as well as having a geographic/location dimension. Ultimately, more complex modelling should be attempted, however this is a good proof of concept. Typically, kNN does not need a train and test split, however we used one since we have complete observations for EIM data, and want to be able to use this much larger database to help inferences in the kelp dataset. After assigning training to EIM, and test to kelp data, we then tested several variations of kNN models: scaled features, unscaled features, and finally used GridSearchCV. GridSearch identified a model that uses “manhattan distance”, with 15 neighbors, and “distance” as the weights.

Model	MSE	R ²	MAE
Scaled Features	9.8137	-1.4931	2.5527
Unscaled Features	6.7819	-0.729	2.153
GridSearchCV	6.1286	-0.5569	2.013

The model performance increases but, overall indicates fairly poor fit to the data. None of the R² values are positive, which means the model is performing worse than simply predicting missing values with the mean temperature, therefore, failing to capture some complexity of the data. The residual graphs are noticeably better, and more evenly distributed around the centerline for the unscaled model as well as model optimized by GridSearch (fig 3). However, the best model was able to predict within about 2°C. Overall, these metrics all indicate that a more sophisticated model that deals with time series or geographic data would be able to capture the variability in the data better. Potentially, doing site specific analysis, or clustering similar kelp bed sites, could increase performance as well.

Comparison of summary statistics for original vs imputed data		
	Original	Imputed
mean	13.4	13.7
std	1.99	0.69
min	9.7	12.8
max	20	15.03

For our purposes, while crude, kNN imputation seems to yield plausible temperatures to fill missing NWSC temperatures. When we look at the distribution of the recorded data vs the imputed data we see that the mean is similar with a smaller standard deviation for the imputed data, so the imputed data is centered around the mean, with less spread and variability than the original data. This makes sense since we trained on EIM data, which has more data, with slightly more spread than the kelp data, and we specifically only imputed eighteen, missing, kelp data points. So we would expect the distribution for those eighteen imputed points to be more conservative estimates in a narrow band of possibilities. Indeed, when we visualize the recorded data and imputed data together we see that imputed data fits the general trends of the data but does not replicate the variability. Specifically, the original data and imputed have a mean of 13.4 and 13.7, respectively; however, original data has a standard deviation of 1.9 to the imputed 0.7, which can be seen in figure 4. kNN does not capture the full complexity of this data, however it is able to produce data that follows the general trends of the original information in a way that makes a reasonable imputation aid for analysis (fig 5). More complex time series or geographic analysis is beyond the scope of the researchers at this time.

Challenges, Future Directions & Conclusions

In conclusion, imputation via kNN provided a reasonable basis for filling in “missing at random” data. By minimizing constraints such as location, time, and depth, and training on supplementary data from the EIM dataset, we are able to provide predicted temperatures for missing records that are within 2°C accuracy. Limitations do exist, however, the kNN approach does not capture the full complexity of the system, although it does provide a framework to produce plausible temperature estimates that align with the general temporal and spatial patterns of the greater region. These challenges expressly warrant further investigation to enhance the predictive power at a

more granular level. With this in mind, we propose this to be a reasonable proof-of-concept for filling missing values in the kelp dataset, but note that kelp forests are subject to highly localized variability. If this variability is to be captured, we will have to take a more complex approach that incorporates kNN as an intermediary step to create subsets of site specific training data for more complex spatio-temporal based regression methods, like Gaussian Regression or GAMs. In the meantime, this lays groundwork for those, more nuanced, modelling attempts, and should serve as a basis for analysis of temperature effects on the health of kelp in the Salish Sea.

Git Repository: https://github.com/ayoung42/5071_Kelp_Temp

https://nbviewer.org/github/ayoung42/5071_Kelp_Temp/blob/main/code/imputation.ipynb (If there are issues viewing the imputation notebook due to folium/ map rendering in GitHub)

Figures:

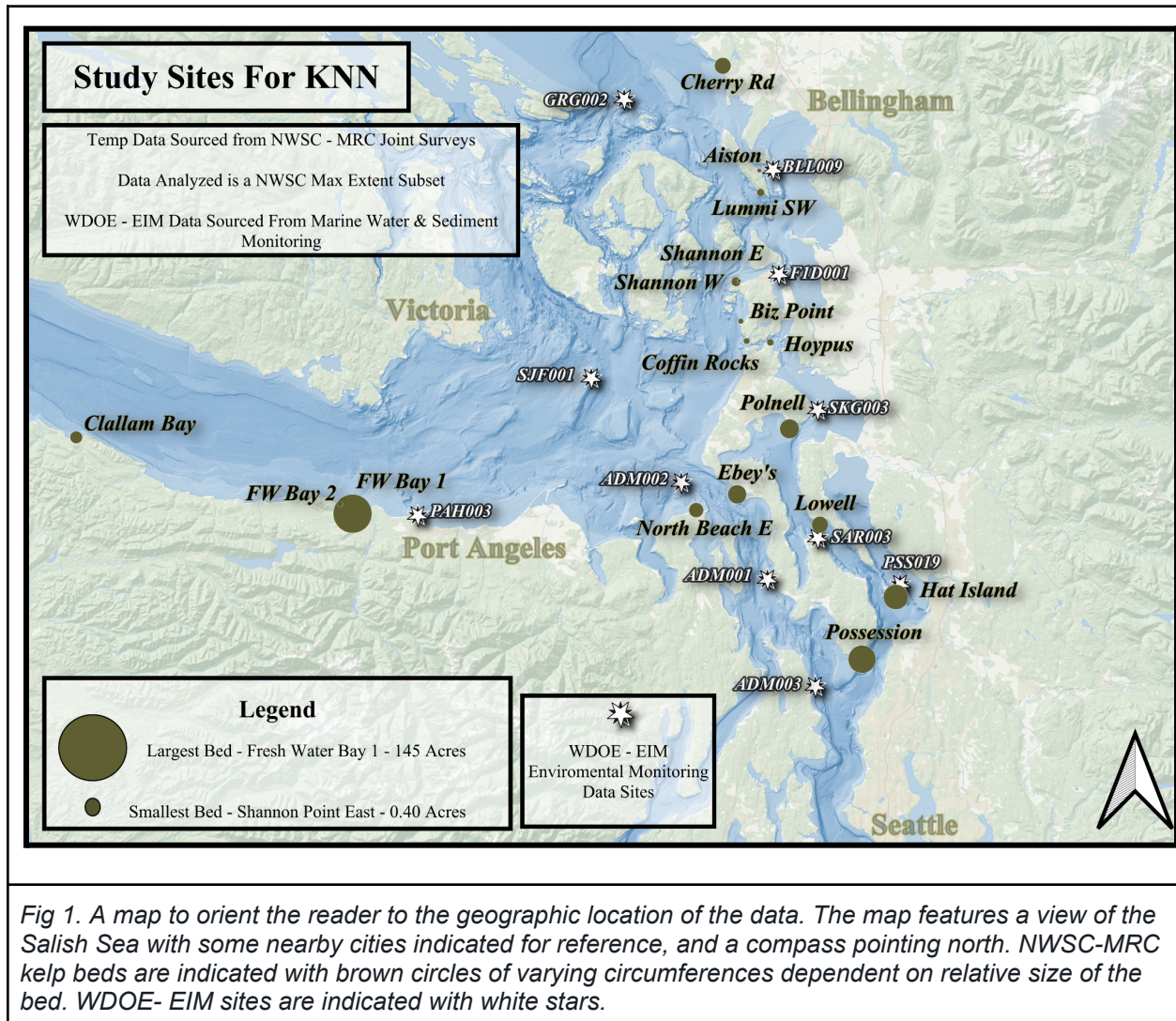


Fig 1. A map to orient the reader to the geographic location of the data. The map features a view of the Salish Sea with some nearby cities indicated for reference, and a compass pointing north. NWSC-MRC kelp beds are indicated with brown circles of varying circumferences dependent on relative size of the bed. WDOE- EIM sites are indicated with white stars.

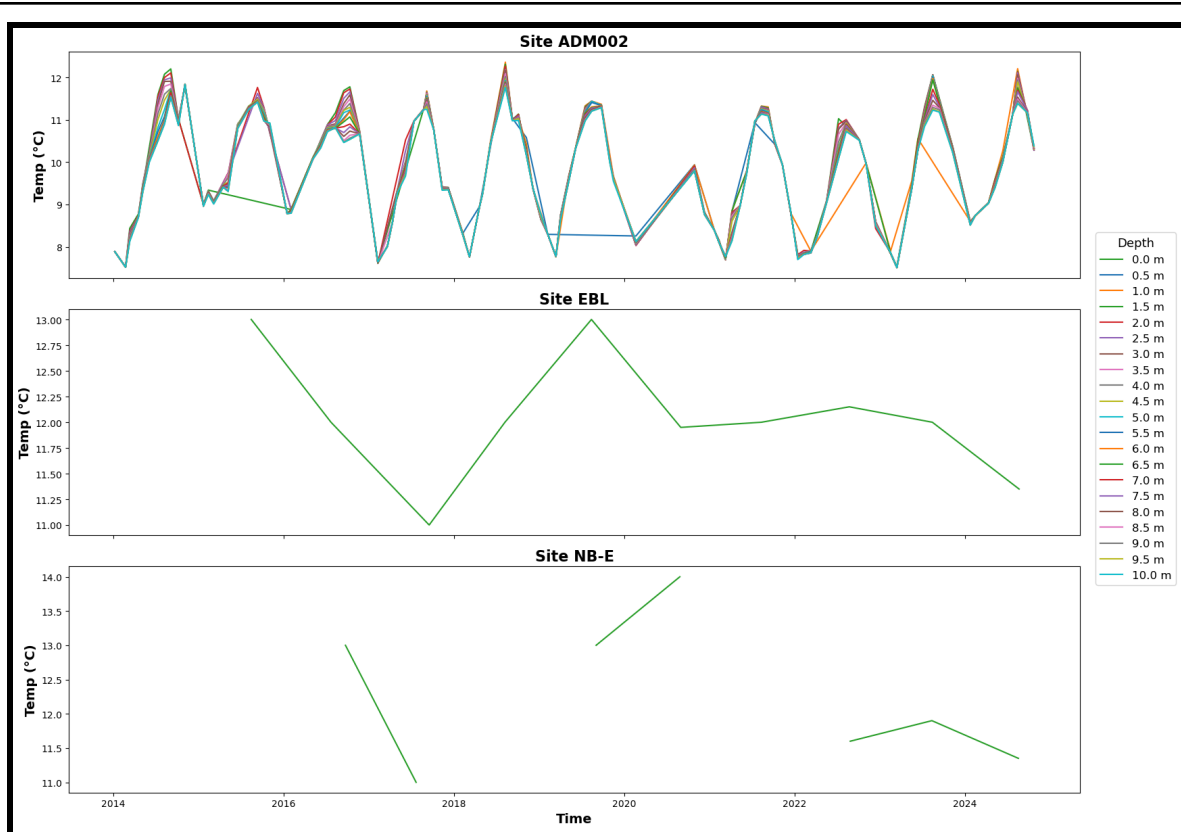
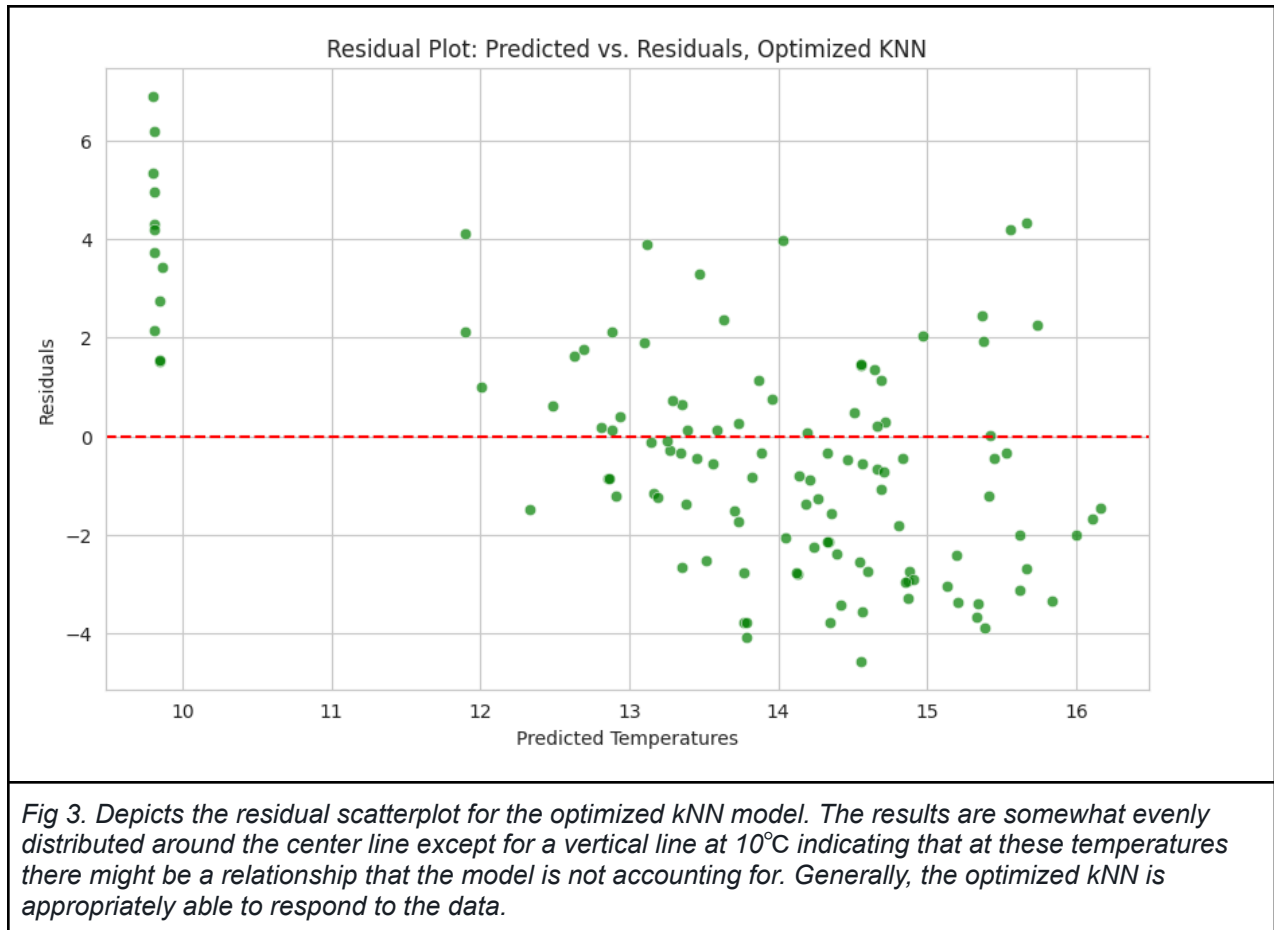


Fig 2. Depicts line graphs of three different locations: ADM002 (a WDOE - EIM site), EBL, and NB-E over time from 2014-2024. ADM002 records data more frequently in comparison to the NWSC datasets in the time frame and has temperatures (8-12°C) at several depths ranging from 0.5-10 meters, but they all follow similar, cyclical, trends. Moving down to the NWSC - MRC dataset, EBL contains data from 2014 onwards, with a depth of 0 meters and temperatures 11-13 °C and will need no imputation. Compare this with NB-E, which has temperature recordings starting in 2017, and is discontinuous to 2024, recorded at 0m and ranging from 11-14°C; this is a great example of a site that needs imputation for missing temperatures. EBL and NB-E were selected to demonstrate a complete dataset, what an incomplete site looks like, and how ADM002 (the nearest WDOE - EIM site to both NB-E & EBL) might serve to fill in some of those blanks.



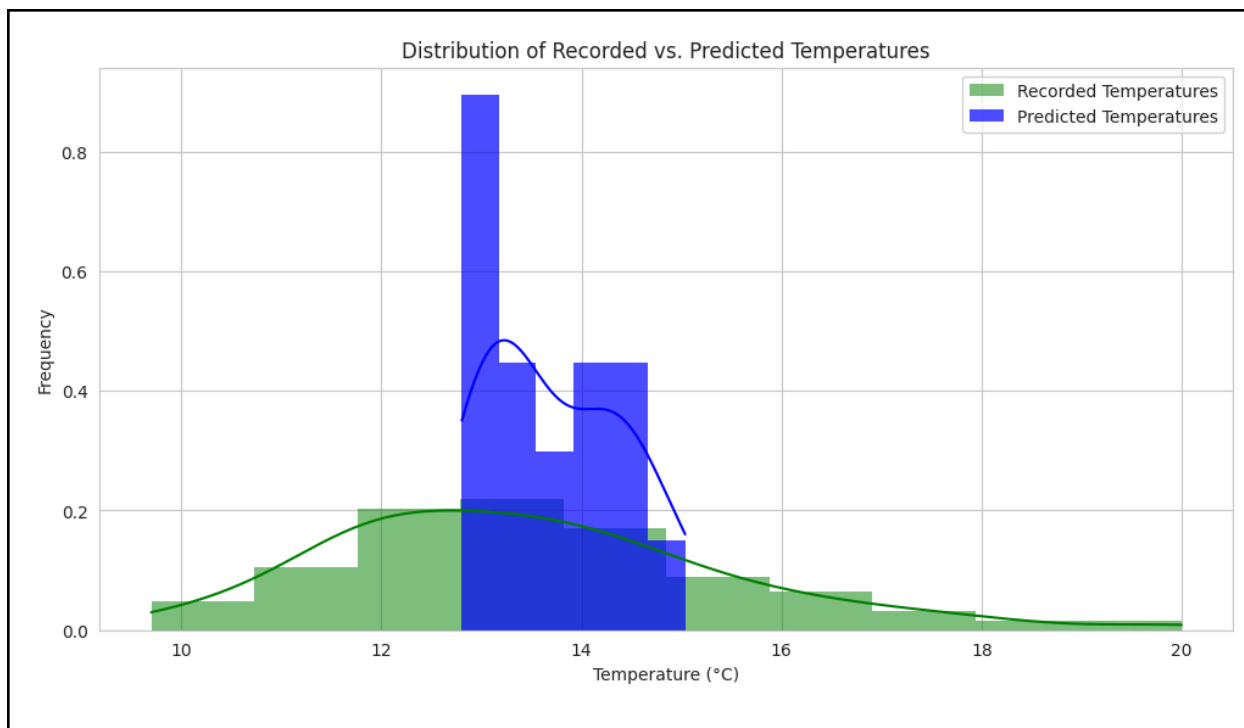


Fig 4. Stacked histograms with recorded temperatures in green and predicted in blue. The recorded temperatures have a distribution from 10-20°C with a mean at about 13 and frequency of 20%. Predicted temperatures have a narrower range of 13-15 and a bimodal peak with the highest mean at about 13.5 with a frequency of 80%. There are less predicted temperatures, and they center roughly around the same point as the original data, as one would expect.

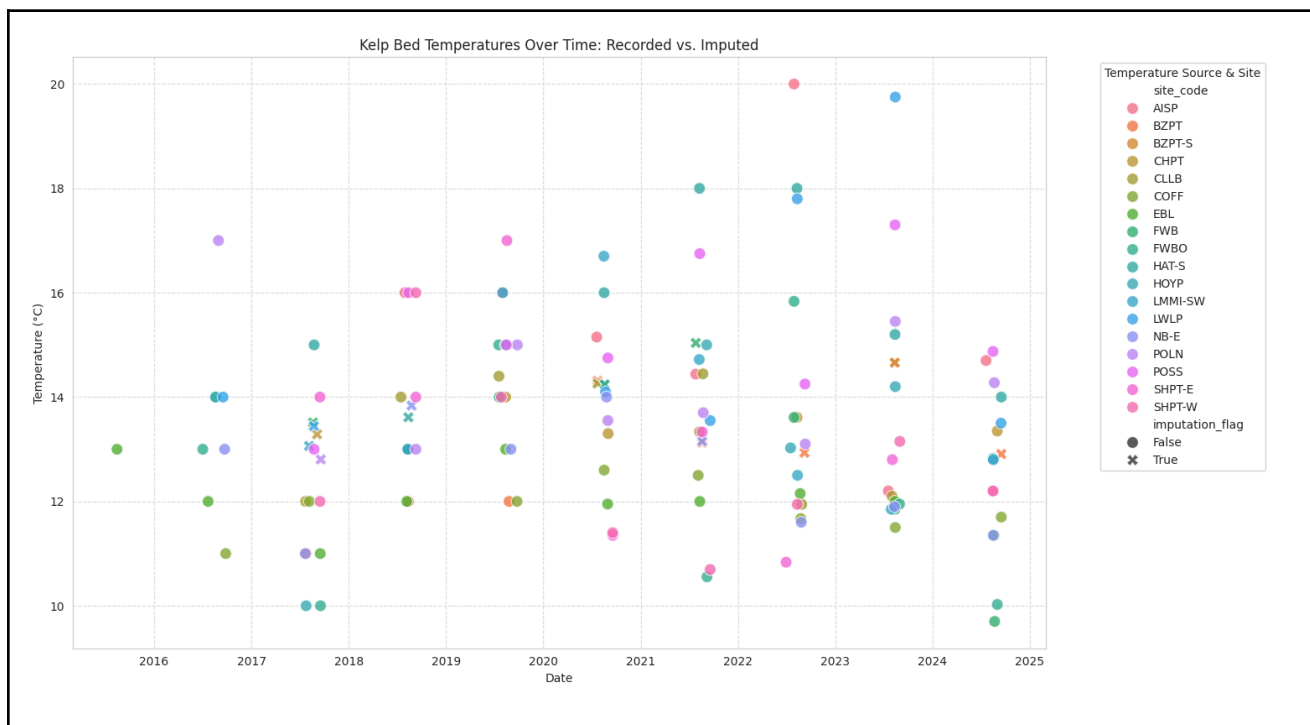


Fig 5. Is a scatterplot of the recorded temperatures and imputed. Each kelp site is color coded and original temperatures are denoted with a "o", while predicted temperatures are marked with an "x". From this we can see that the imputed temperatures follow the same general trends as the recorded data.

References:

- [1] Webb, C. - Whitty, J. (2025). Northwest Straits Commission, Marine Resource Committees kayak kelp monitoring project dataset (2016–2024) [Unpublished raw data, not for publication]. Personal communication. A subset of this data can be found on SoundIQ: <https://www.nwstraits.org/our-work/soundiq/>
- [2] Washington State Department of Ecology (2026). Environmental Information Management database [Environmental monitoring data portal]. Washington State Department of Ecology. A searchable database of environmental monitoring data including air, water, soil, sediment, and biological sampling. Available at: <https://ecology.wa.gov/research-data/data-resources/environmental-information-management-database> (accessed Feb 14, 2026).
- [3] Drakard et al., "Impact of Multiple Climate Stressors on Early Life Stages of North Pacific Kelp Species." <https://onlinelibrary.wiley.com/doi/10.1002/ece3.71661>
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- [5] Weigel et al., "Effects of Temperature and Nutrients on Microscopic Stages of the Bull Kelp (*Nereocystis Luetkeana*, Phaeophyceae)." <https://onlinelibrary.wiley.com/doi/10.1111/jpy.13366>